

# CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

## University Examination

Date: 24/05/2018

Day: Thursday

Time: 10:00am to 10:30am

Total Marks: 20

|                           |  |  |  |  |  |  |  |  |  |
|---------------------------|--|--|--|--|--|--|--|--|--|
| Student's ID No.          |  |  |  |  |  |  |  |  |  |
|                           |  |  |  |  |  |  |  |  |  |
| Student's Name: _____     |  |  |  |  |  |  |  |  |  |
| Student's Sign : _____    |  |  |  |  |  |  |  |  |  |
| Supervisor's Name : _____ |  |  |  |  |  |  |  |  |  |
| Supervisor's Sign: _____  |  |  |  |  |  |  |  |  |  |

|                                                                                          |
|------------------------------------------------------------------------------------------|
| Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following |
| Examination : BOP- May 2018                                                              |
| Semester: III                                                                            |
| Section : I                                                                              |
| Course Code : BOP303                                                                     |
| Course Name: Optometric Optics I                                                         |

### Important Instructions:

- There are two sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes. Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

### SECTION-I

**Multiple Choice Questions: (MCQs)**  
(Encircle the correct answer)

(10 × 1 = 10)

1. Fresnel prism can be manufactured by a process of\_\_\_\_
  - a. Injection molding
  - b. Compression molding
  - c. Both A&B
  - d. NOTA
2. If 3 prism diopters BU and 1 prism diopter BU is placed in front of RE and LE respectively, then what will be the resultant vertical prismatic effect?
  - a. 2 PDBU, OD
  - b. 2 PDBD, OS
  - c. 4 PDBU, OD
  - d. 4 PDBD, OS

3. **Barrel shaped distortion is caused by**
- Positive lens
  - Glass slab
  - Minus lens
  - Prism
4. **Spherical equivalent for prescription of +2.00/+5.75×90 is\_\_**
- +4.75Ds
  - +4.87Ds
  - +3.87Ds
  - +3.75Ds
5. **Which of the following is correct?**
- Distance power + Near power = Near Rx
  - Near power – Distance power = Near addition
  - Distance power – Near power = Near addition
  - Near power – Distance power = Near Rx
6. **Cornea is of \_\_\_\_ form.**
- Plano convex
  - Bi convex
  - Bi concave
  - Meniscus
7. **Power of lens with focal length 15cm is\_\_**
- 6.36D
  - 6.66D
  - 6.00D
  - 6.99D
8. **A lens clock is calibrated for glass of index \_\_\_\_\_**
- 1.525
  - 1.530
  - 1.535
  - 1.515
9. **What does the root word Phot means?**
- Light
  - Color
  - Image
  - Disease
10. **From following colors which one is having shortest wave length?**
- Orange
  - Red
  - Yellow
  - Blue

**Identify the sentence as True or False:**

**(Please tick mark the answer)**

**(5 × 1 = 5)**

**True    False**

**11.** Cross curve is always weaker/flatter than the base curve.

|                          |                          |
|--------------------------|--------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> |
|--------------------------|--------------------------|

**12.** Higher the Abbe value, the more chromatic aberration present in the lens.

|                          |                          |
|--------------------------|--------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> |
|--------------------------|--------------------------|

**13.** Fresnel prism is made up of PVC material.

|                          |                          |
|--------------------------|--------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> |
|--------------------------|--------------------------|

**14.** 1 degree is equal to 1.75 prism diopters.

|                          |                          |
|--------------------------|--------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> |
|--------------------------|--------------------------|

**15.** Minus lens is thinner at centre than in the periphery.

|                          |                          |
|--------------------------|--------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> |
|--------------------------|--------------------------|

**Match the following**

**(5 × 1 = 5)**

|                                                 |  |                               |
|-------------------------------------------------|--|-------------------------------|
| <b>16.</b> Neutralizing power                   |  | <b>A.</b> Dispersion          |
| <b>17.</b> VIBGYOR                              |  | <b>B.</b> Front surface power |
| <b>18.</b> 1.525                                |  | <b>C.</b> Fresnel prism       |
| <b>19.</b> Measuring phoria & fusional vergence |  | <b>D.</b> Distortion          |
| <b>20.</b> Petzval's surface                    |  | <b>E.</b> Back surface power  |
|                                                 |  | <b>F.</b> Risley prism        |
|                                                 |  | <b>G.</b> Curvature of Image  |

(16 -    ),    (17-    ),    (18-    ),    (19-    ),    (20-    )

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

## III Semester of BOP Examination

May 2018

### BOP303 Optometric Optics I

Date: 24/05/2018

Time: 10:30am to 1:00pm

Maximum Marks: 50

#### Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

#### SECTION-II

##### Q.1. Short answers (10 out of 11)

(10 × 2 = 20)

- a. Convert 2 cylinder form into optical cross (1) +4.00×20, -3.00×110, (2) -1.00×100, -11.00×10
- b. A certain lens of index 1.523 has a convex spherical surface. The sag of the front surface is 1mm for a chord whose length is 20mm. what is power of front surface?
- c. Explain Base curve and Cross curve.
- d. A +6.50D lens before the right eye is decentered 3mm nasalward. What amount of prism is induced and what is the base orientation?
- e. Write down formula of front and back vertex power.
- f. Explain about Resultant horizontal prismatic effects.
- g. Describe spherical lens form.
- h. Convert spherocylindrical form into optical cross (1)+2.00/+1.00×180 (2)-4.00/-3.00×90
- i. What is longitudinal chromatic aberration?
- j. Define Equivalent power.
- k. Define (1) Prism (2) Refracting angle of prism

##### Q.2. Short Notes (4 out of 5)

(4 × 5 = 20)

- a. +2.00/+3.00×180, Base curve=+6.00. Find out the Fraction value using toric transposition steps.
- b. Explain in detail about spherical equivalent with example.
- c. Derive Sag formula with proper diagram.
- d. Write a note on Prentice's rule.
- e. Explain in detail about spherical aberration

##### Q.3. Essay (1 out of 2)

(1 × 10 = 10)

- a. Fresnel Press on prism
- b. A lens has the following parameters. F1=+10.00D, F2= -7.00D, n=1.580, tp=1.0mm, lens diameter=50mm. find out tc using exact sagitta formula.